Inferring hybridisation

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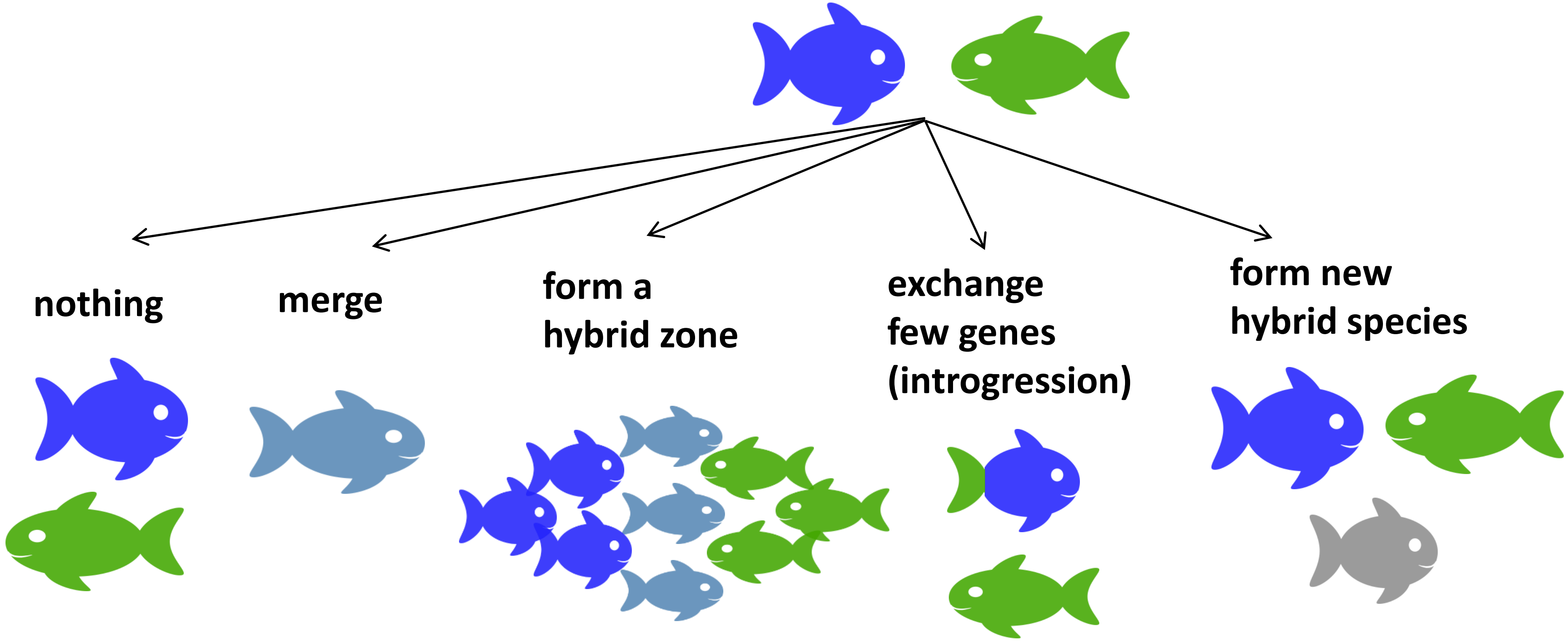
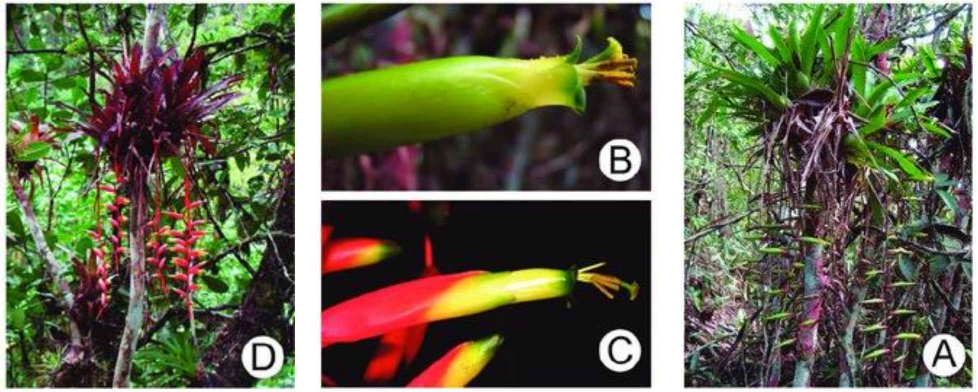
Evolutionary consequences of hybridisation



Hybridisation at different time and divergence scales

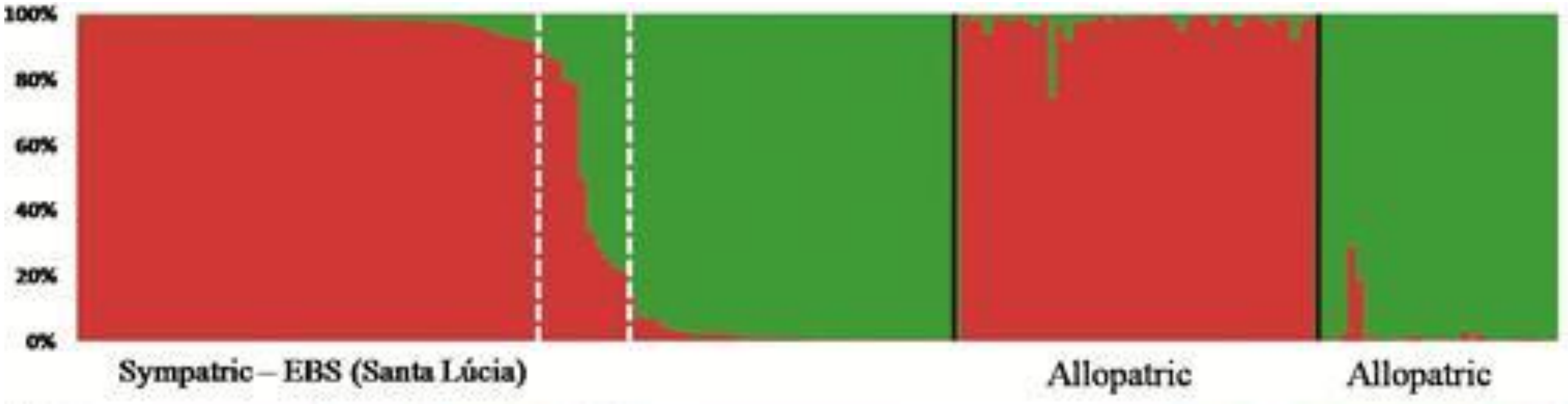
- **Current ongoing hybridisation:**
  - PCA to get a first overview
  - STRUCTURE/ADMIXTURE/fastSTRUCTURE to get a general overview and infer admixture proportions for each individual
  - Triangle plots to classify hybrids
  - Potentially cline analyses if the hybrids are geographically restricted

Example of clustering methods to identify hybrids and infer ancestry proportions: e.g. bromeliads



Vriesia simplex *Vriesia scalaris*

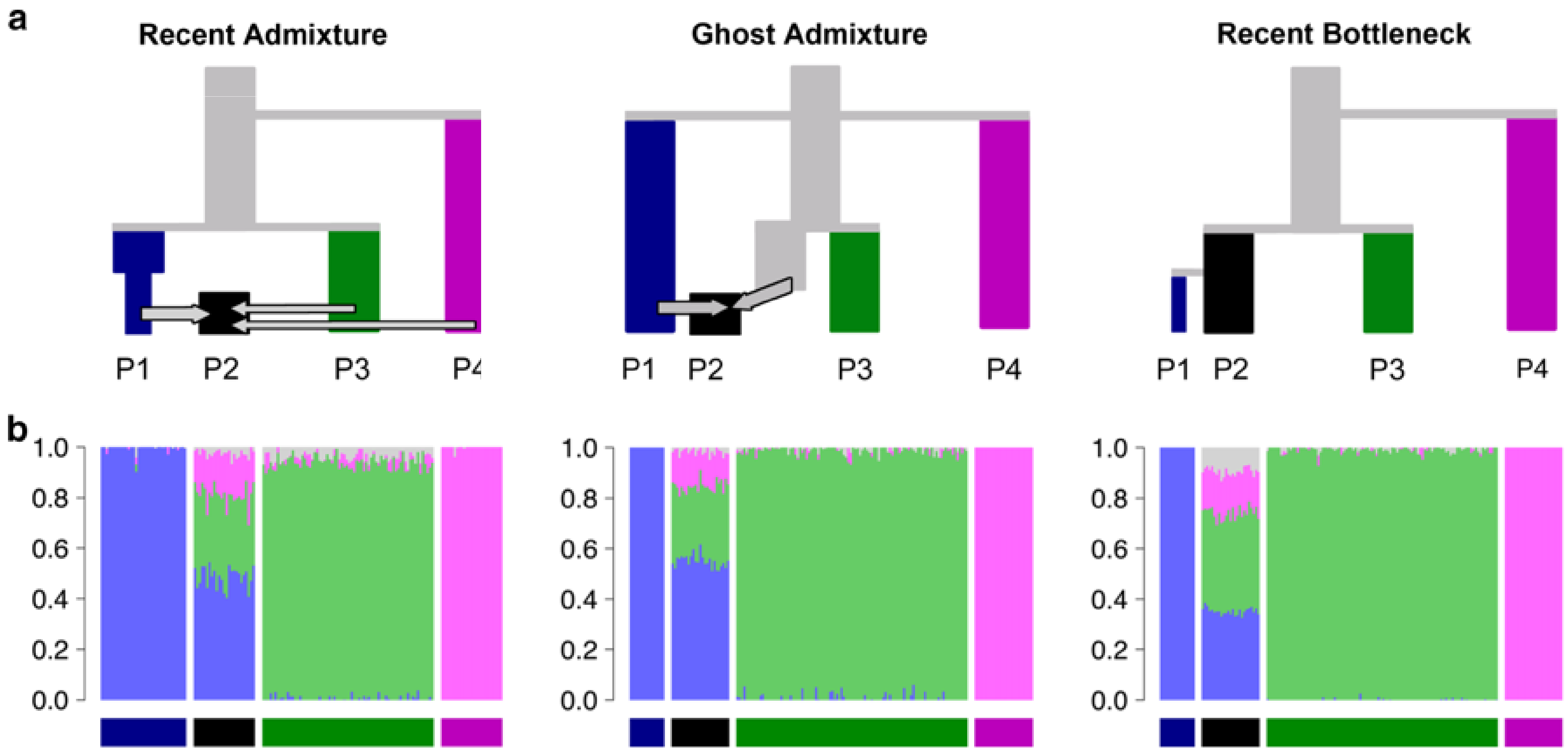
Ancestry proportion



Tutorial: https://speciationgenomics.github.io/ADMIXTURE

Neri, et al., AoB Plants

Very different evolutionary histories can give the same ADMIXTURE/STRUCTURE plot

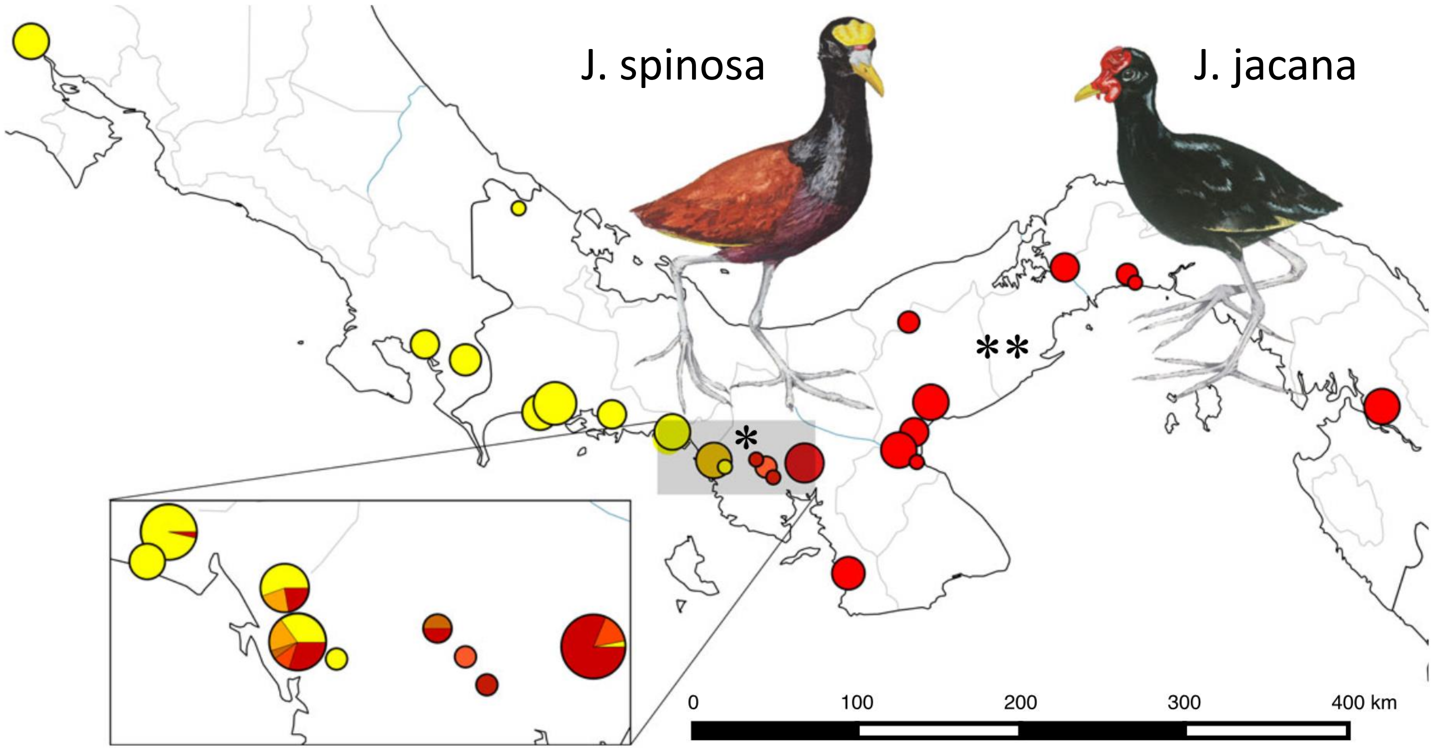


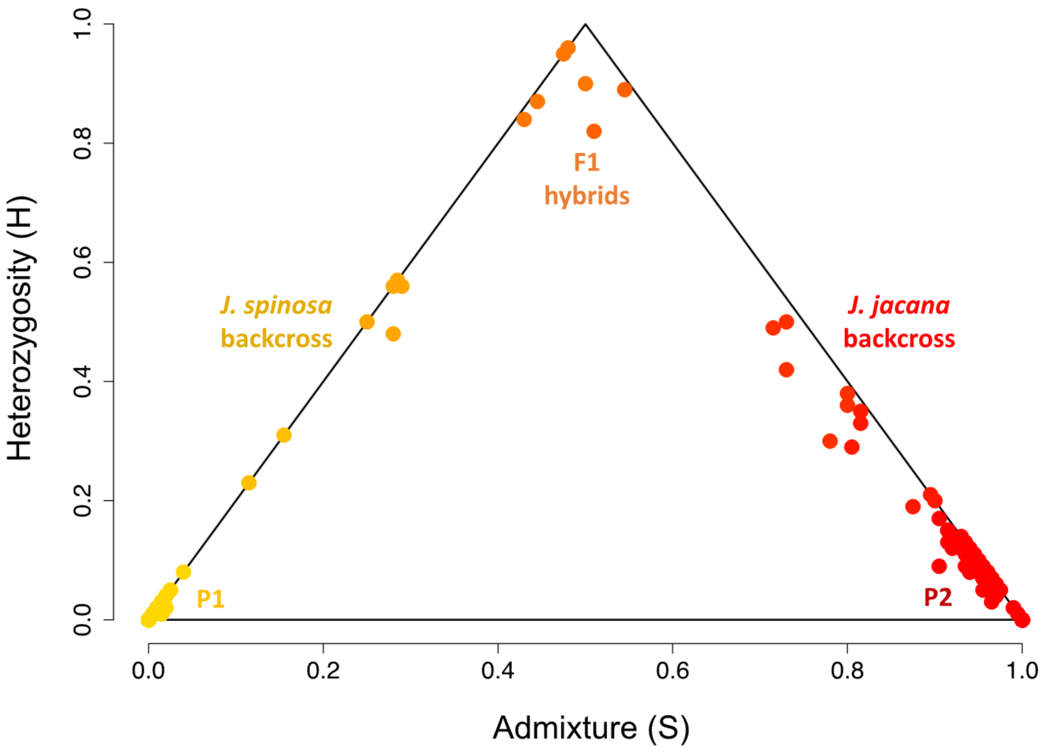
Very important: Use STRUCTURE/ADMIXTURE plots only to get a first overview of genetic structure. Putative cases of hybridization need to be followed up with other analyses such as D statistics and/or demographic modeling. Potentially use 'badMixture' (by Dan Lawson) to check how well your result fits the data.

Lawson et al., 2018, NatComm

Triangle plots help to identify recent hybridisation (e.g. triangulaR)

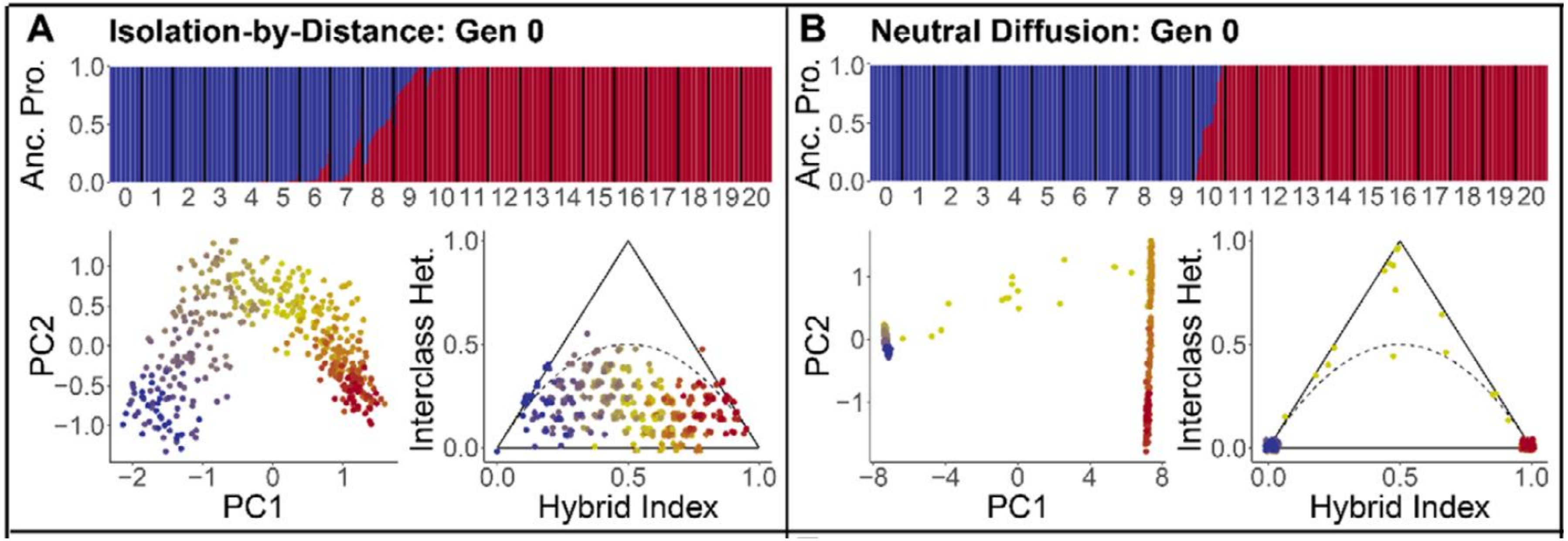
Triangle plot shows assymmetric introgression





Lipshutz et al. 2019

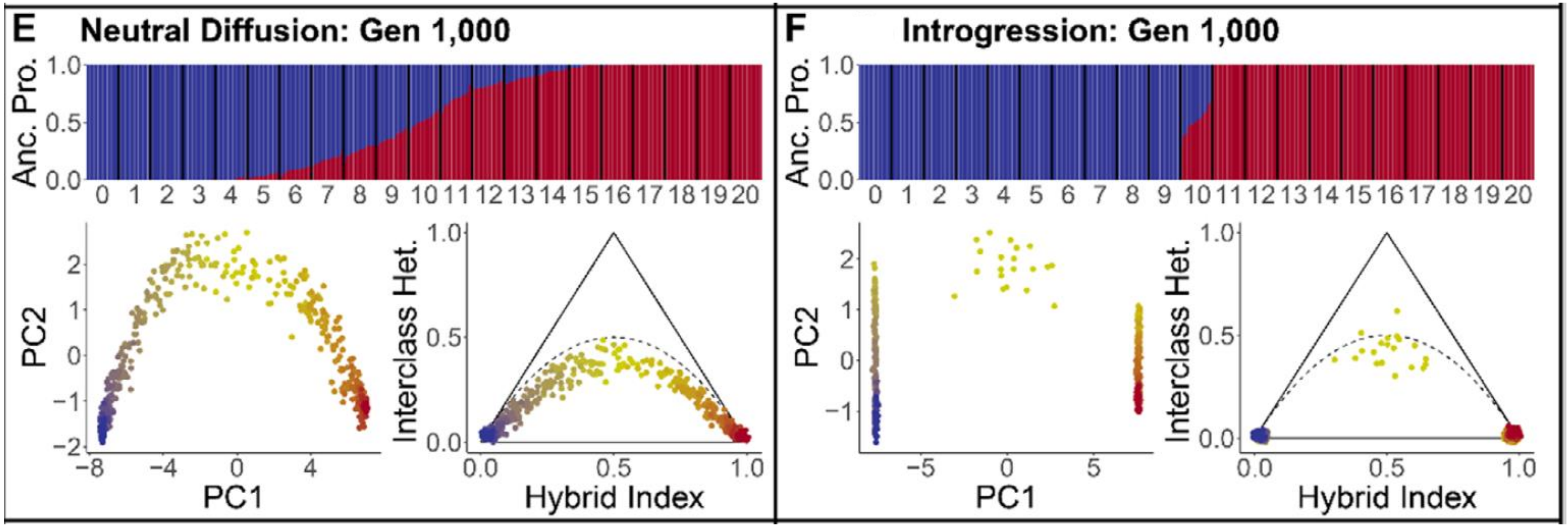
Also strong isolation by distance can mimic hybridisation in STUCTURE plots -> use triangle plots (e.g. triangulaR)



Wiens & Colella, 2024

Isolation by distance can mimic hybridisation in STUCTURE plots -> use triangle plots (e.g. triangulaR)

1000 generations after the initial contact: we get all kinds of hybrids if there is no reproductive isolation or hybrids restricted to a small hybrid zone if there is reproductive isolation.



Wiens & Colella, 2024

conStruct (Bradburd, Coop, and Ralph 2018) allows distinguishing isolation by distance from introgression

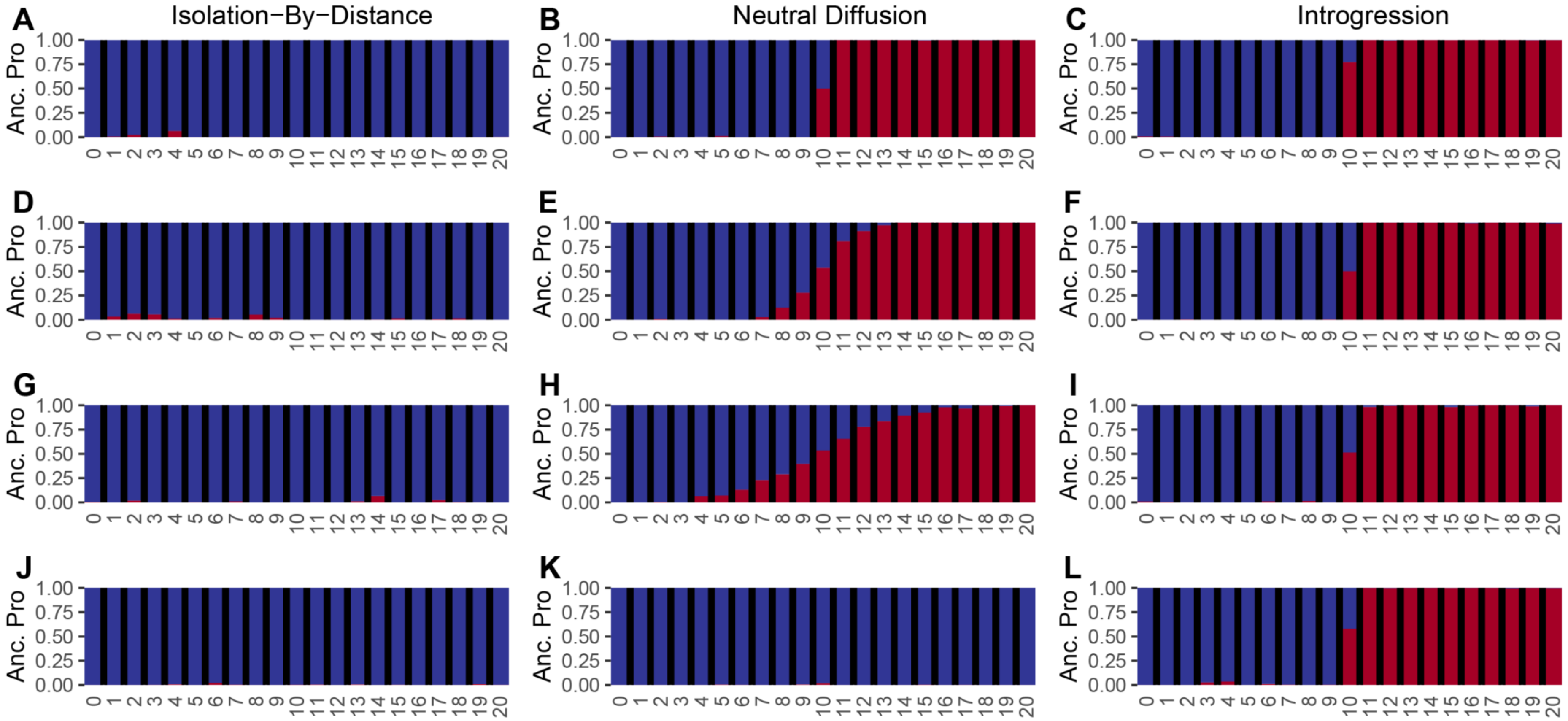
generation

0

200

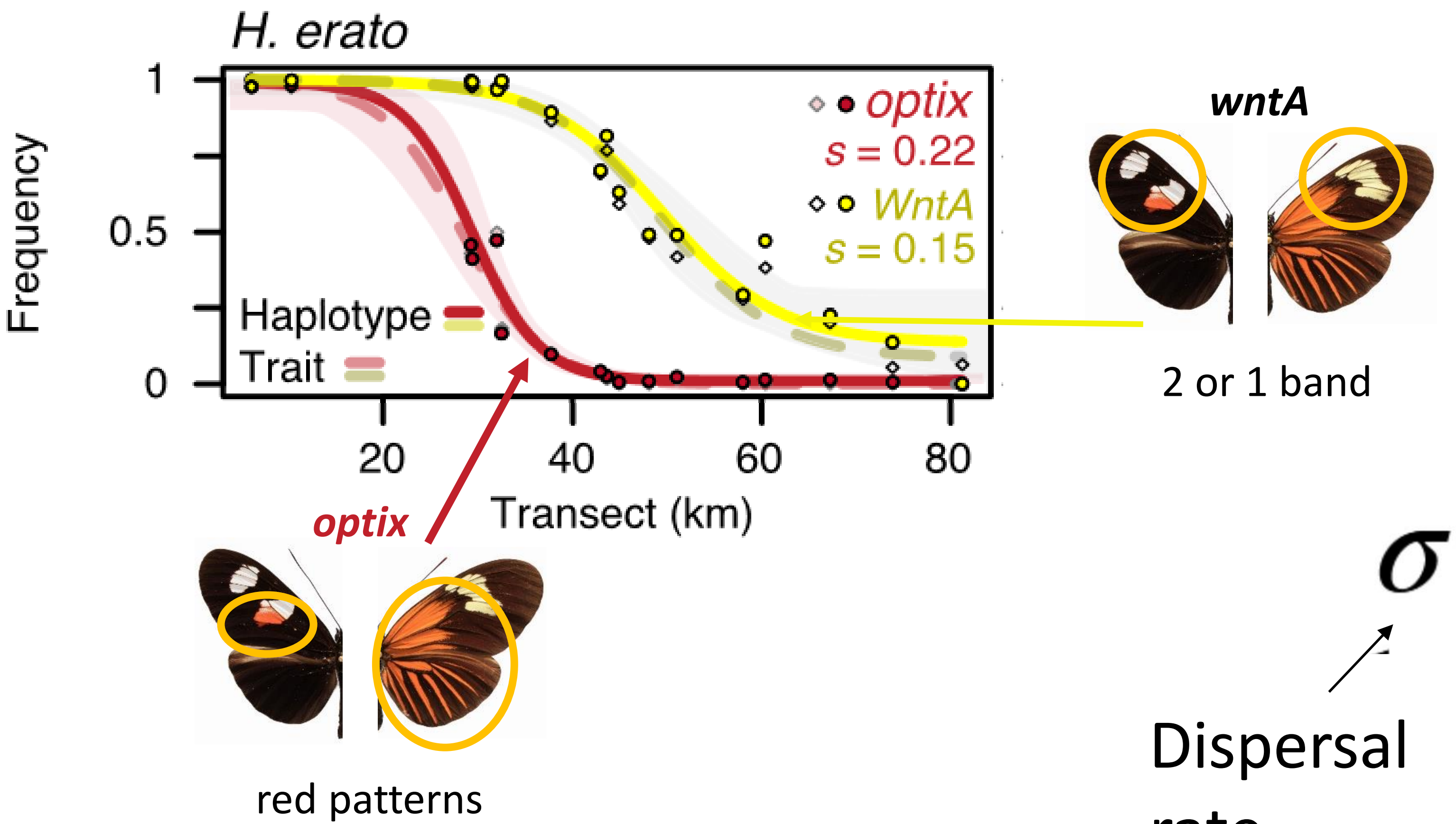
1000

10000



http://www.genescape.org/construct.html

Hybrid zones allow us to test for barriers to reproductive isolation and estimate selection



$$\sigma \sim w\sqrt{s/12}$$

Dispersal rate

Cline width

Selection coefficient

Hybridisation at different time and divergence scales

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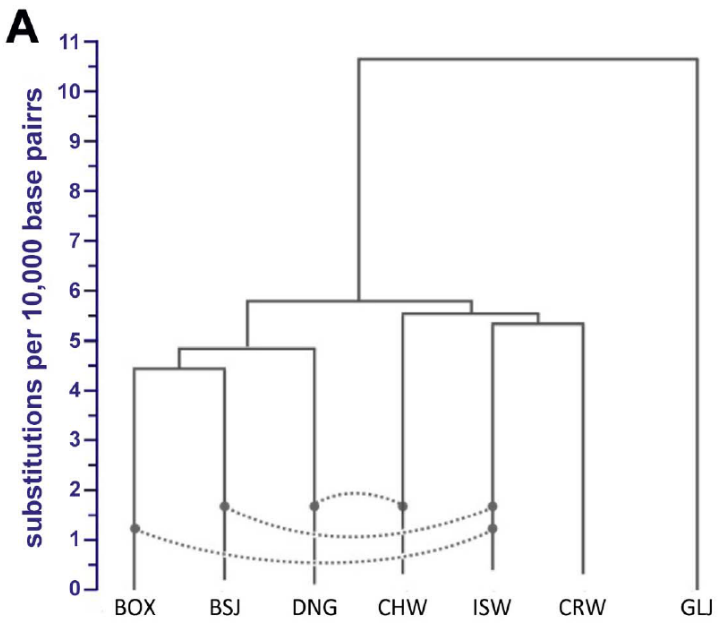
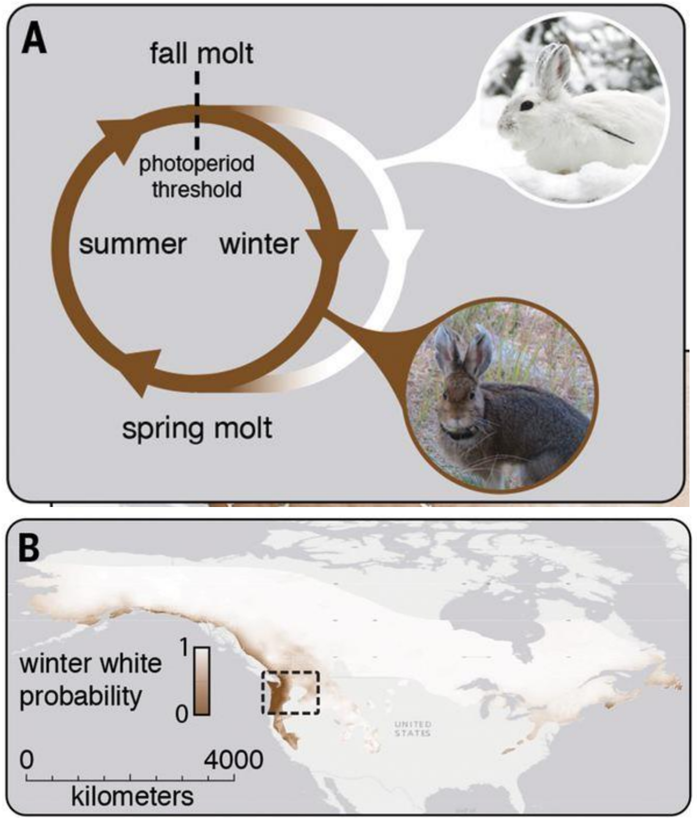
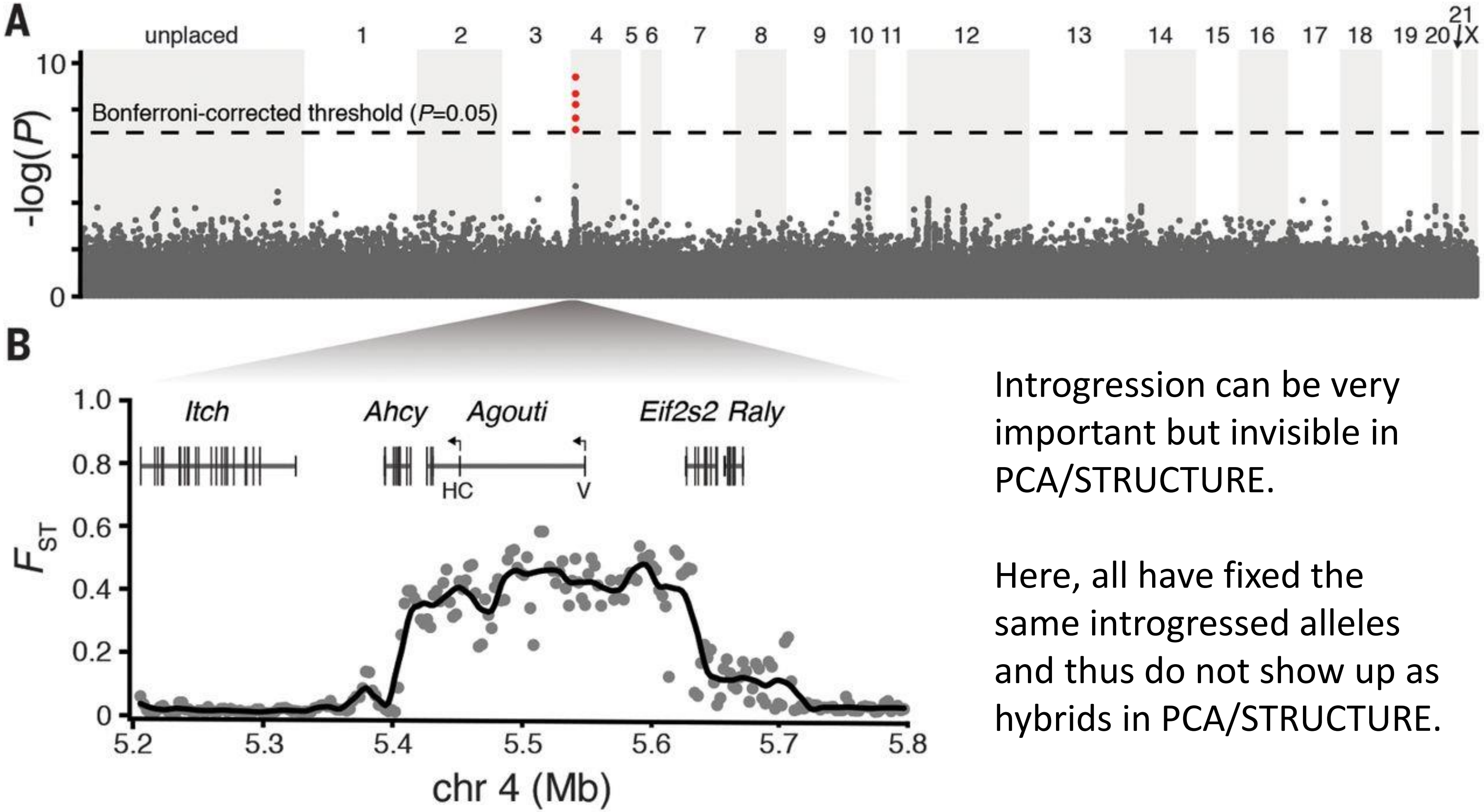
- **Past hybridisation (introgression, potentially between distant relatives)**
  - Gene flow between sister species/groups:
    - Demographic modelling (e.g. fastsimcoal)
    - FST/Dxy
    - fd scans if unadmixed sister population/species is available
  - Between more distant relatives:
    - D statistics (=ABBA/BABA test), F branch
    - Phylogenetic methods that include introgression (e.g. PhyloNet, SpeciesNetwork, AIM, GPhoCS, treemix, etc)
    - Comparing phylogenies at different loci (e.g. TWISST)



Example of adaptation to climate change: Adaptive introgression in snowshoe hares



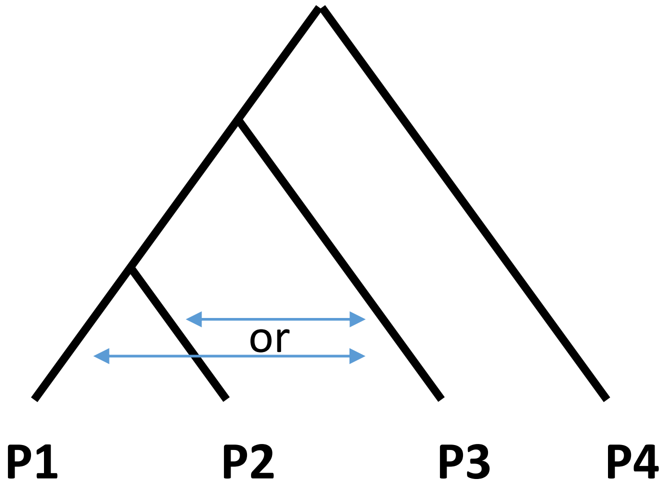


Introgression can be very important but invisible in PCA/STRUCTURE.

Here, all have fixed the same introgressed alleles and thus do not show up as hybrids in PCA/STRUCTURE.

Jones et al. 2018

Patterson’s D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let’s imagine these are our sequences (haploid for simplicity)

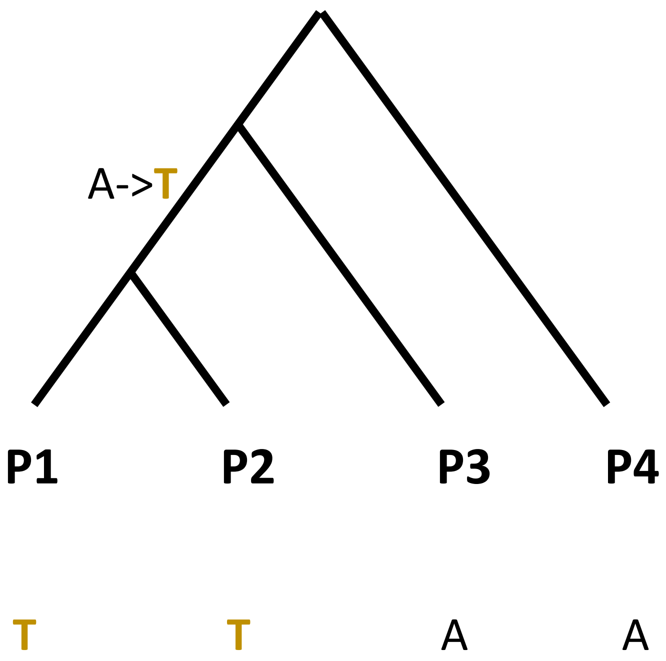
P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

P4: A---T---T---A---A-- (outgroup)

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: **T**---G---C---C---A--

P2: **T**---G---T---A---T--

P3: **A**---G---T---C---T--

P4: **A**---T---T---A---A-- (outgroup)

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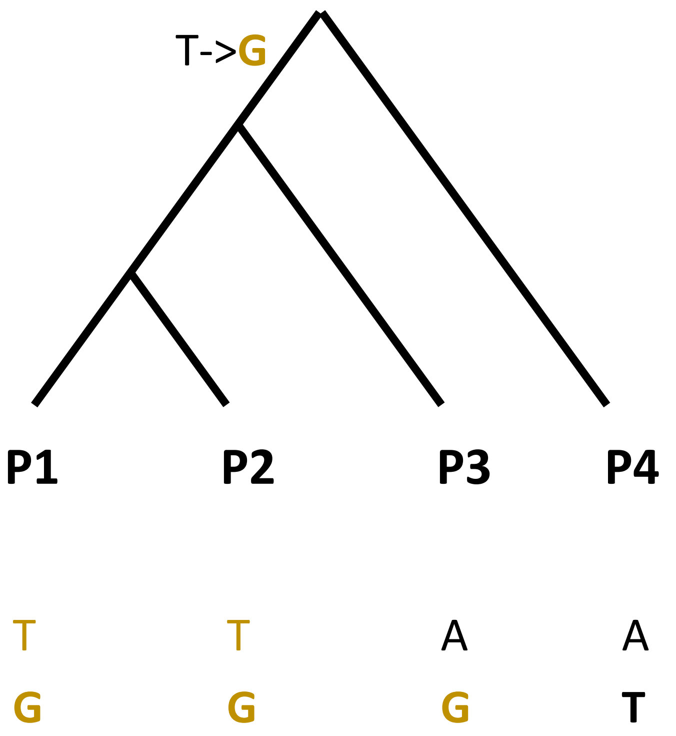


Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

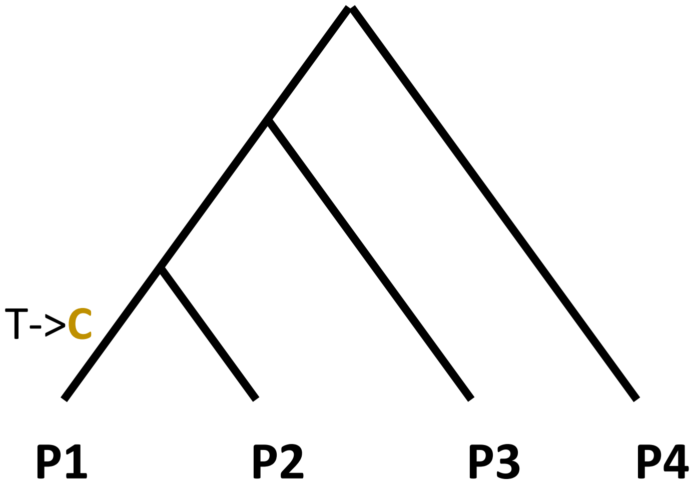
P1: T---**G**---C---C---A--

P2: T---**G**---T---A---T--

P3: A---**G**---T---C---T--

P4: A---**T**---T---A---A-- (outgroup)

Patterson’s D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let’s imagine these are our sequences

P1: T---G---**C**---C---A--

P2: T---G---**T**---A---T--

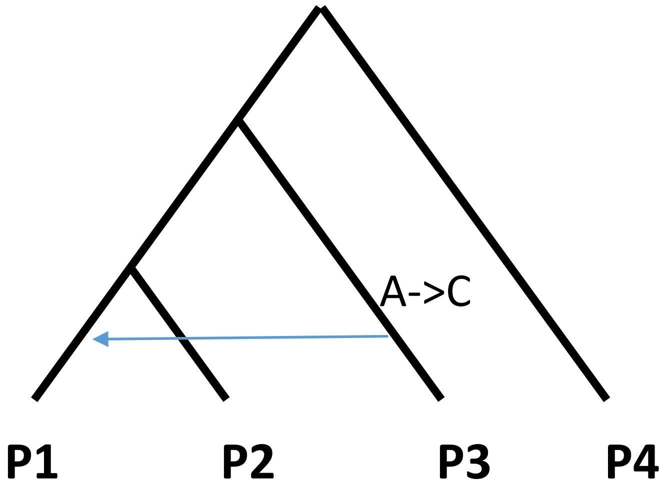
P3: A---G---**T**---C---T--

P4: A---T---**T**---A---A-- (outgroup)

T	T	A	A
G	G	G	T
C	**T**	**T**	**T**

Concordant SNPs

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---**C**---A--

P2: T---G---T---**A**---T--

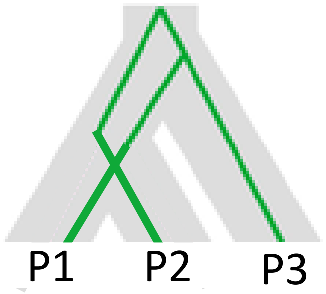
P3: A---G---T---**C**---T--

P4: A---T---T---**A**---A-- (outgroup)

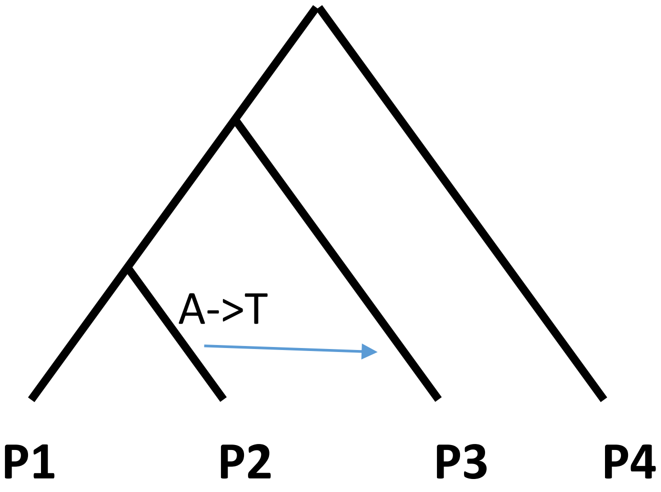
T	T	A	A
G	G	G	T
C	T	T	T
C	**A**	**C**	**A**

Concordant SNPs

Could also be due to incomplete Lineage sorting



Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---**A**--

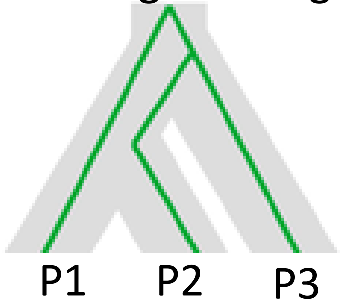
P2: T---G---T---A---**T**--

P3: A---G---T---C---**T**--

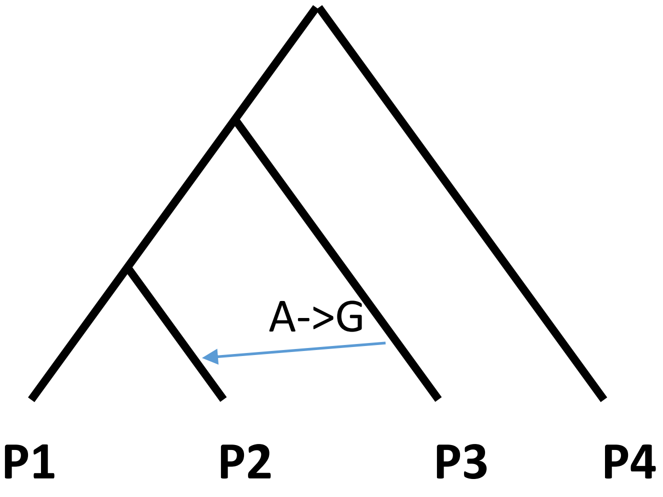
P4: A---T---T---A---**A**-- (outgroup)

P1	P2	P3	P4	
T	T	A	A	Concordant SNPs
G	G	G	T	
C	T	T	T	
C	A	C	A	Discordant SNPs
A	**T**	**T**	**A**	

Could also be due to incomplete Lineage sorting



Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---A--**A**-

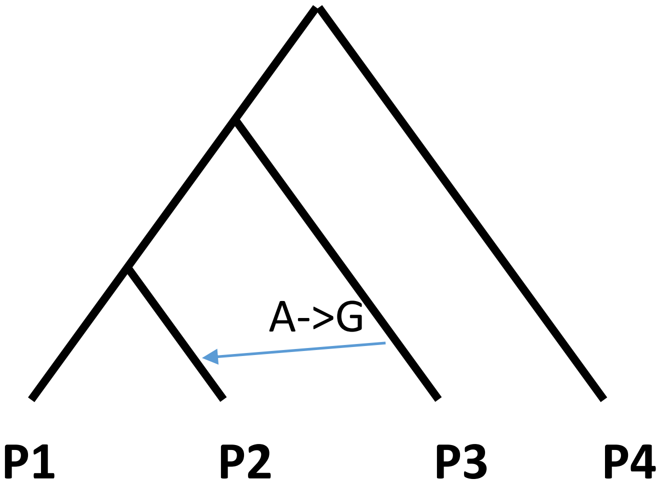
P2: T---G---T---A---T--**G**-

P3: A---G---T---C---T--**G**-

P4: A---T---T---A---A--**A**- (outgroup)

T	T	A	A	Concordant SNPs
G	G	G	T	
C	T	T	T	
C	A	C	A	Discordant SNPs
A	T	T	A	
A	**G**	**G**	**A**	

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---A--A-

P2: T---G---T---A---T--G-

P3: A---G---T---C---T--G-

P4: A---T---T---A---A--A- (outgroup)

T	T	A	A	Concordant SNPs
G	G	G	T	
C	T	T	T	
C	A	C	A	BABA
A	T	T	A	ABBA
A	**G**	**G**	A	ABBA

$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$$D = \frac{2-1}{2+1} = 1/3$$

Supplementary

Model assumptions of ADMIXTURE/STRUCTURE

- SNPs are unlinked -> LD-pruning is very important
- Individuals are unrelated -> Do not include siblings
- Populations are represented by similar number of individuals -> strongly underrepresented populations may not be detected
- Ideally use at least 10,000 SNPs
- Sites are bi-allelic and singletons are removed

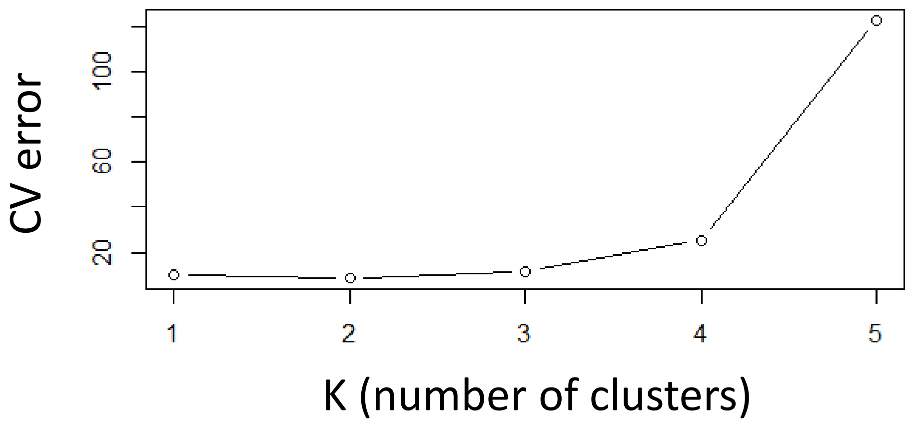
Limitations of ADMIXTURE and similar methods

ADMIXTURE and similar methods are great at inferring how many groups of individuals the dataset contains and which ones belong to which group. However, they struggle with the following:

- Detection of clusters that are strongly undersampled (e.g. just a single very divergent individual may not be picked up)
- Detecting ancestral admixture that is shared by all individuals of a cluster
- Handling individuals that are highly inbred or with loads of missing data

How to identify the best number of clusters?

ADMIXTURE uses a cross-validation (CV) procedure:
e.g. 5-fold CV: the genotypes are randomly split into five partitions. Each partition in turn is masked (set to missing) and then predicted with the individual cluster assignment and allele frequencies inferred for each cluster. The predicted genotypes are compared to the real genotypes to infer the prediction error. The prediction errors together are used to infer the cross validation error.
The lowest CV error indicates the best number of K but don't just use the CV error.



Make sure to always look at the plots with different numbers of clusters and also use your knowledge on the biology and collection places to identify which number of clusters is most informative.

STRUCTURE vs ADMIXTURE

- STRUCTURE is a model-based clustering approach which utilizes genotype data to infer the presence of distinct populations and assign individuals to populations. The number of populations (k) has to be predefined by the user. Usually, one tests multiple values and checks at which k the model posterior probability start to level out. Using an MCMC approach, STRUCTURE identifies k clusters that are each in HWE and assigns individuals to these clusters or a mix of them.
- ADMIXTURE works similarly but uses a maximum likelihood estimation to identify clusters and infer individual ancestries. The best value of k is determined with cross-validation.
- ADMIXTURE is much faster than STRUCTURE and thus ideal for large NGS datasets. STRUCTURE may be better suited for identification of very weak population structure also because it can use prior probabilities for the cluster assignments. However, ADMIXTURE allows to specification of unadmixed reference individuals with known ancestry.

Is the ADMIXTURE plot a good fit for your data?

- **badmixture**
  (Dan Lawson)

- **Evaladmix**
  (Garcia-Erill & Albrechtsen)

  Positive correlation of residuals (red) indicates that the individuals in this group are more closely related than ADMIXTURE suggests, blue means that they are less closely related than modelled by ADMIXTURE.